Semantic Sensor Web and IoT for Agricultural applications

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Abstract. Internet of Things play an indispensable role in our day to day lives by monitoring different aspects of our surroundings and enhancing the quality of life through smart ways. There is an increase in the volume and heterogeneity of the data generated every moment. To ensure interoperability and seamless overlay of data from across domains, and to develop applications which can capture real-world abstractions from the raw data, there is a need for machine interpretable semantics. Semantics enable situation awareness and contextual information in IoT and allows machines to understand and reason on the data. To combine the semantics of heterogeneous Geospatial data, we need to have well structured and unambiguously defined ontologies. A scalable and reusable semantic middleware for developing IoT applications is discussed. This study focuses on cross-domain applications of IoT data and design of domain and application-specific Ontologies. A semantics-driven support system for agricultural decision making is developed, which combines data from various IoT devices, using the ontologies and performs reasoning to derive inferences on disease diagnosis on a specific horticulture crop.

Keywords: Internet of things · Ontologies · Semantic Sensor Web · Agriculture · Decision Support Systems

1 Introduction

1.1

Internet of Things enables objects embedded with sensors that can observe their surroundings to capture some information and also use actuators to trigger an action based on that sensed data. Mobile phones, Personal Assistant Agents, Smart watches are just a few of the numerous IoT devices. Smarter, smaller and cost-effective IoT devices and rapid analysis for event detection is the need of the day. The data from the equipment and other sources have to be gathered and

processed in an effective manner to improve their processes, predict and prevent problems, and ultimately create superior ecosystems [1].

For overall monitoring and decision support systems, applications across domains must be able to discover the desirable IoT devices, and retrieve information without a-priori knowledge about the sensors - their networks or processes. Standardization is the key to interoperability. Sensor Web Enablement activities proposed by the Open Geospatial Consortium specifies standardized data models and service models. The XML based models provide for syntactical and structural interoperability. It opens up the sensors and sensor data for easy discovery and access. However, huge volumes of data are generated every second from these connected devices. Semantic approaches, which enhances the context of the IoT data need to be used for efficient detection and retrieval of significant events from the voluminous data.

Ontology is a method of formally defining the concepts and relationships in a domain. The logic of the real world concepts should be incorporated in the sensor data for meaningful analysis and knowledge retrieval.

1.2 Semantic Sensor Web

Semantic web, as envisioned by Berners-Lee et al. [2] is an extension to the Web, in which meaning of the information is well defined. In the World Wide Web, the documents are linked to form a web of pages. But it is designed for human end-user. In the semantic web, the meaning of the data or metadata is of significance. Understanding the meaning of the data helps in its effective use. The semantic web provides for unique identification of all resources and connects them with relationships. Context and logic is instilled into the data, thus enabling machines to understand and interpret the content. It supports machine to machine effective communication and knowledge-based search. The main component of semantic enrichment is the formulation of domain knowledge in the form of ontology.

We use the domain of Agriculture as a motivating application to describe the need and applicability of Semantic Sensor web. Towards this we propose an approach for integrating information from IoT devices and use it to predict the succeptibility to a certain type of disease in Citrus Orchards. F To achieve this the his expert knowledge is put in the form of ontologies in semantic-based

systems to enable the machines to understand and derive information, through logical reasoning .

The core objectives are:

- Design an ontology using W3C SSN, comprising of general concepts of IoT device observations and measurements
- Extend the base ontology to develop domain-specific ontologies and rules for agriculture and plant pathology
- Develop a semantic framework to build cross-domain IoT applications

2 Related Work

The idea of Semantic Web and its potential in the world of sensors led to plentiful research in the area of ontologies, implementation of the domain knowledge, linking the sensor data to existing ontologies Linked Data approaches, etc. Semantic Sensor Web enhances the meaning of sensor observations and provides better situational awareness and accurate real-time knowledge, thereby aids in emergency situations like disasters [3]. Sensor data is annotated with spatial, temporal and thematic information using XML based Resource Description Framework (RDF). Web Ontology Language(OWL) is used to define the domain concepts and relationships. The knowledge base thus formed with RDF triples and OWL ontologies is used to derive new ontological assertions using rule-based reasoning tools like Semantic Web Rule Language.

Similarly, with semantic annotation using O&M in OWL, a Semantic Sensor Observation Service (SemSOS) is implemented using 52 North, which enables complex querying over simple weather data. Gray et al. describes a semantic sensor web architecture for integrating multiple heterogeneous datasets, including live and historic sensor data, databases, and map layers used for Environmental Decision Support Applications [5]. Several existing ontologies - domain-specific, infrastructure and interoperability related - are linked to form an ontology network.

Fernandez et al. describes the semantic sensor web architecture for developing autonomous agents for intelligent transport network [6]. Sensor sub domain used the W3C SSN-XG ontology. Sensor data is annotated with SSN ontology concepts and stored as RDF triples. Agent layer consists of multi-agents for different scenarios like Driver Personal Agent, Traffic Light Agent, Road Agent etc.

Kim et al. designs an ontology-based context modeling for an intelligent service in an agricultural farm [7]. Crop growth knowledge and the Farm environment parameters are the main concepts in the ontology. The design of

ontology requires developing class models (concepts), individuals and properties between them.

Wang et al. describes the significance of converting the vast knowledge present in the books to computer resources. This is necessary for integration, sharing of data and for semantic interoperability [8]. This paper focuses on compiling hilly citrus production knowledge into ontologies.

Decision making is influenced by a variety of parameters across domains. Creating cross-domain applications is necessary to model the real life scenarios. In the context of the Internet of Things, there is a need to explicitly describe the meaning of the Machine to Machine (M2M) data provided by M2M devices. Gyrard et al. describes a Semantic-based Machine-to-Machine Measurement(M3) approach to automatically combine, enrich and reason about M2M data to provide cross-domain applications [9]. M3 ontology acts as a hub to fuse the cross domain knowledge.

This study proposes to design a base ontology for easy integration of different domain specific ontologies. Methodology for developing IoT applications using the Semantic approach is formulated. This is demonstrated by designing a semantics-driven agricultural decision support application.

3 Methodology

This study is focused on developing frameworks to gain insights and deduce suggestions from Geospatial IoT data. The applications should aggregate the IoT data, detect the events and notify the users or take an action

3.1 Motivating IoT Application for Citrus Orchards

An important application area of IoT is Agriculture/Horticulture. The farmer's experience and knowledge is perhaps the most determining factor for the activities in a farm. A semantic approach can derive insights from the raw sensor data and provide answers to queries of the user such as :

Which areas of the farm do crops have water stress? How much irrigation should be done? Are any crops affected by some disease? What are the remedial measures?

These are some of the questions that can be solved using semantic approaches because. The various environmental and crop parameters are considered by farmers to decide on the activities like irrigation, pest control, fertilization. A scientific approach for monitoring and providing inputs to farmers will help in better maintaining the crops, increasing productivity and promoting sustainable use of resources.

An ontology based method will help in retrieving the relevant events from the raw sensor data. The knowledge base needs to be created with inputs from the farm experts, scientists and the vast knowledge in the books, which defines the crops and requirements scientifically. Thus, context-aware application can be built, which will aid in agricultural decision support.

By studying the inherent characteristics and using semantic approaches, it will be able to predict and therefore prevent the possible occurrence of problems

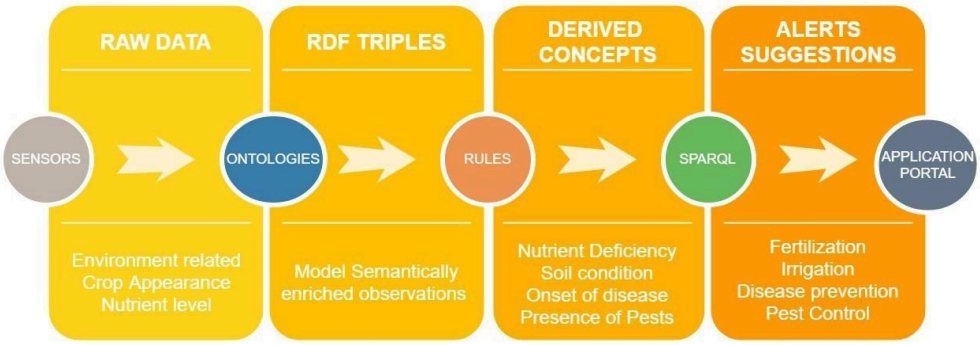


Fig.1. Overview of the application for agriculture

like diseases. Using information about the type of soil, the moisture content and expert knowledge, we can precisely give inputs on irrigation measures. Planning for crop cultivation can be done as per the latest scientific updates. Weather monitoring in the areas can predict any adverse conditions and suggest precautionary measures. Sensors can also be used for surveillance in the farms and storage areas.

3.2 Semantic Framework for IoT Application

The architecture of the proposed IoT application for agriculture is shown in Figure 2. The Ontology layer, reasoning layer and querying layer together form the Semantic Middleware, between the sensor data and the user application layer. It converts the sensor data to meaningful semantically rich knowledge.

Data acquisition layer comprising of the Things of the IoT are the sensors which collect the data. SenseQube is an interoperable sensing system deployed to measure the agro-meteorological parameters [10]. It provides real time data of environmental parameters - Temperature, Humidity, Soil Moisture, Rainfall, Wind

speed. Data acquired by SenseQube sensors and observations is the primary input to the semantic application.

Ontology layer formally defines the data, makes it more structured, logical and imparts the meaning to the IoT data. The terminology related to IoT sensing, observations, samples are explicitly specified in the ontology. The pattern of parameters like temperature, soil moisture, nutrient content can be studied by measuring them using sensors. The SSN ontology is used to semantically annotate the sensors and their observations, enabling it to be used in building RDF graph models. The domain ontologies formally define the application specific or domain related concepts.

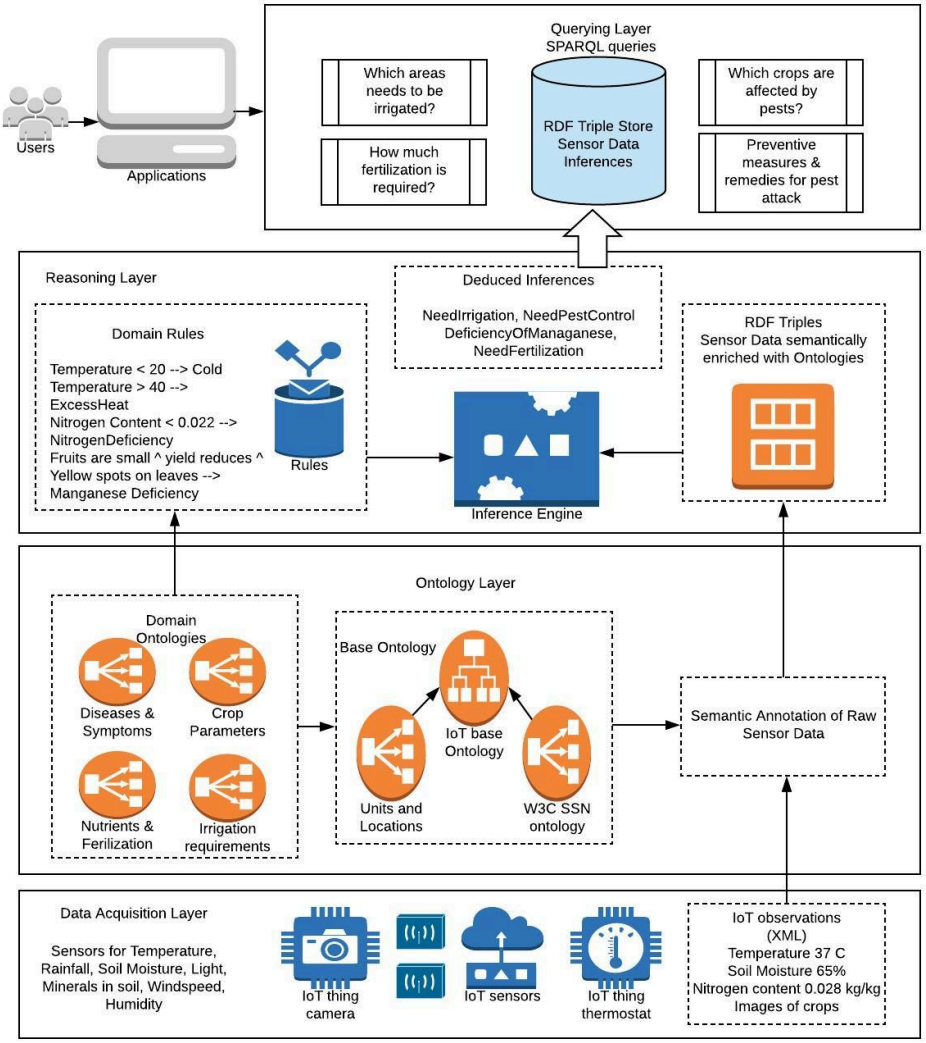


Fig.2. Proposed Architecture of Semantic Driven application for Agriculture

Reasoning Layer Along with ontologies, the rules determining the high level events are formulated. The rules help in identifying what the IoT data implies, or the real world significance of the specific data. The RDF data, ontologies, rules together forms a knowledge base. Reasoners and rule based inference engines

can reason on the knowledge base to derive inferences, detect anomalies and deduce more knowledge.

Querying Layer Using ontologies, the raw data will be converted to subject-predicate-object RDF triples and can be used to build RDF models. They can be stored in RDF Triple store. The semantically annotated IoT data, ontologies and inferences together form the knowledge base. Querying is done on this knowledge base to retrieve inferences, recommendations etc. Application specific or end user specific SPARQL queries are formulated for fetching the required information.

4 Implementation and Results

4.1 Semantics-Driven Application for Agriculture

The aim of this study is to develop semantic middle-ware for Internet of Things applications for agriculture. SenseQube, an Agro-meteorological IoT platform for Agriculture is deployed in farms and provides the weather and soil parameter values [10]. An agricultural decision support application is developed which uses semantic approach on sensor data for disease prediction in crops.

4.2 Ontologies

Sensors SSN and SOSA ontology is imported for the sensor concepts. Protege [11] was used for developing the ontologies. Figure 3 shows the major concepts and their relations in the SOSA ontology, which are used in this implementation visualized in VOWL viewer.

Plant Diseases Ontology for plant protection [12] incorporates the plant diseases domain knowledge. The PPO ontology was extended for Citrus plants. The classes in this ontology is shown in Figure 4. It relates plant disorders with the observed abnormalities. The abnormalities included are appearance, lesions or color. This application focuses on Citrus plant diseases caused by Phytophthora spp. The domain information about the diseases, symptomatology, epidemiology is obtained from scientific papers related to Citrus Diseases [13]. To formulate the plant disease knowledge, different instances of fungal diseases, their corresponding symptoms are defined in the ontology.

Semantic Sensor Network for Agriculture (SSNAGRI) Importing the SSN and PPO ontology,the proposed Semantic Sensor Network for Agriculture (SSNAGRI) was developed. It consists of concepts related to crop, location,

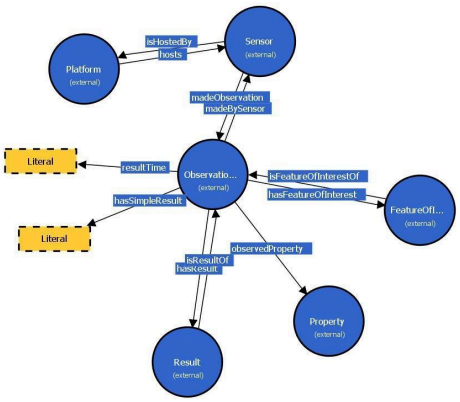


Fig.3. SOSA/SSN ontology

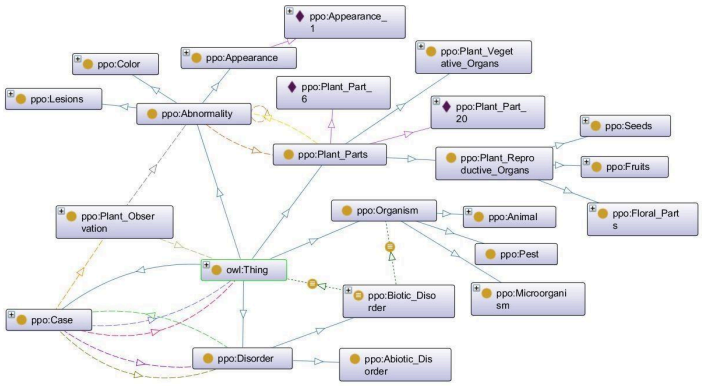


Fig.4. Classes in PPO ontology

environmental parameters, disease indices, remedies etc. All the crop related classes along with their object and data properties is visualized using VOWL as

shown in Figure 5. To model the scenario, various individuals are created for the classes.

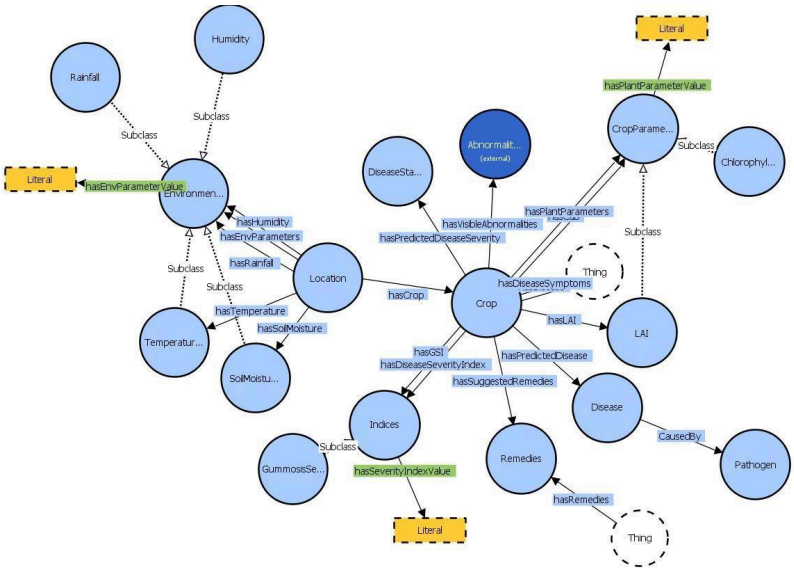


Fig.5. Crop and Diseases Ontology

4.3 Rules

Rules for the following events detection are formulated using Semantic Web Rule Language (SWRL) in Protege.

- Gummosis Severity Index estimation from the weather and plant parameters
- Prediction of Disease and Severity from Severity Index
- Disease derivation from plant appearance observations
- Environment parameter values from sensor values

Disease Severity Index Model Gummosis Severity index is calculated using weather parameters and plant biophysical parameters using the model in Equation 1 [14], where T is the temperature, R is Rainfall, LAI is leaf area index and Cab denotes Chlorophyll content. Corresponding SWRL rule is in Figure 6.

Crop biophysical parameters like Chlorophyll and Leaf Area Index is obtained using LUT based approaches for Inverse PROSAIL model [14].

```
Rule: Gummosis Severity Index from Parameters
ssnagri:Crop(?cr) ^ ssnagri:isLocatedIn(?cr, ?loc) ^
ssnagri:hasTemperature(?loc, ?t) ^ ssnagri:hasRainfall(?loc, ?r) ^
ssnagri:hasCab(?cr, ?c) ^ ssnagri:hasLAI(?cr, ?l) ^ ssnagri:hasGSI(?cr, ?g)
^ ssnagri:Temperature(?t) ^ ssnagri:Rainfall(?r) ^ ssnagri:Chlorophyll(?c)
^ ssnagri:LAI(?l) ^ ssnagri:hasEnvParameterValue(?t, ?t1) ^
swrlb:multiply(?t2, ?t1, -0.0666) ^ ssnagri:hasEnvParameterValue(?r, ?r1) ^
swrlb:multiply(?r2, -0.0033, ?r1) ^ ssnagri:hasPlantParameterValue(?l, ?l1)
^ swrlb:multiply(?l2, -0.0148, ?l1) ^ ssnagri:hasPlantParameterValue(?c,
?c1) ^ swrlb:multiply(?c2, 0.2134, ?c1) ^ swrlb:add(?gs, ?t2, ?r2, ?l2,
?c2, 3.2631) -> ssnagri:hasSeverityIndexValue(?g, ?gs)
```

Fig.6. SWRL Rule for Gummosis Severity Index

$$GS = -0.0666 * T - 0.0033 * R - 0.0148 * LAI + 0.2134 * Cab + 3.2631 \qquad (1)$$

Disease occurrence and severity [15] are also predicted using Gummosis Severity Index using SWRL Rules.

Plant Disease from Observed Symptoms The crops are instantiated using concepts from SSNAGRI. Each disease and its observed plant changes are formulated in the PPO ontology. When crops indicate some symptoms, the corresponding disease needs to be identified. The rule for the same in shown in Figure 7.

```
Rule: Disease Identification from symptoms
ssnagri:Crop(?c) ^ ssnagri:hasVisibleAbnormalities(?c, ?viab) ^
ppo:Plant_Observation(?po) ^ ppo:observed_abnormality(?po, ?obsab) ^
ppo:Case(?case) ^ ppo:has_plant_observation(?case, ?po) ^
ppo:case_of(?case, ?d) ^ sameAs(?viab, ?obsab) ^ ssnagri:hasRemedies(?d,
?rem) -> ssnagri:hasDisease(?c, ?d) ^ ssnagri:hasSuggestedRemedies(?c,
?rem)
```

Fig.7. SWRL Rule for Disease Identification from Plant Observations

4.4 Reasoning and Querying

Pellet reasoner engine in Protege was used to validate the ontology and rules and the inferences were verified. The probable diseases, severity, symptoms, remedies for a Crop are inferred from the environmental parameters of the crop location, the plant parameters and visible abnormalities.

Snap SPARQL query plug-in was used to query the inferred model and retrieve the derived information. Few examples of questions that can be answered by this model are listed below. SPARQL queries formulated for each of the listed questions and the results are in the following figures.

1. Which crops are affected by diseases? (Figure 8)
2. There are some unusual changes in the bark. What are the possible diseases? (Figure 9)
3. What are the different diseases caused by Phytophthora? What are the symptoms and the remedies? (Figure 10)
4. For a location, which are the different sensors deployed and the observations recorded. (Figure 11)

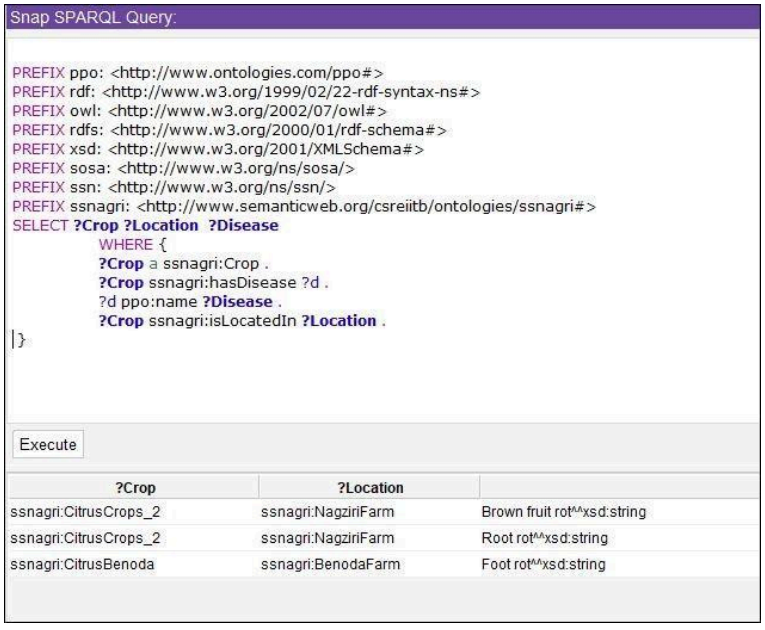


Fig.8. SPARQL Query in Protege

4.5 Semantic Application Framework

The first step is the conversion of the raw sensor values to semantically annotated RDF format. The sensor data in RDF form are then used to create models using Apache Jena. Sensor and Domain ontologies are added to the model. GenericRuleReasoner is used to perform reasoning and inferences. SPARQL queries

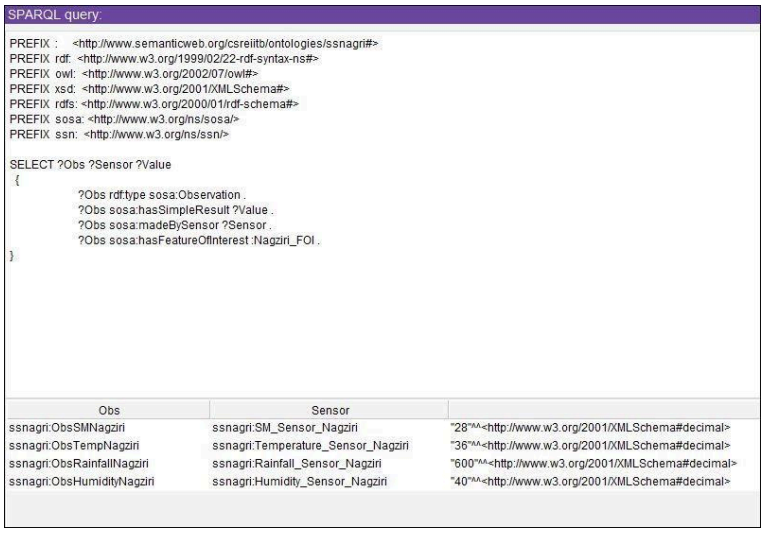


Fig.9. SPARQL query for sensors observing Nagziri

Snap SPARQL Query:

```
PREFIX rdf: <http://www.w3.org/1999/02/22-rdf-syntax-ns#>
PREFIX owl: <http://www.w3.org/2002/07/owl#>
PREFIX rdfs: <http://www.w3.org/2000/01/rdf-schema#>
PREFIX xsd: <http://www.w3.org/2001/XMLSchema#>
PREFIX ppo: <http://www.ontologies.com/ppo#>
SELECT ?disease ?causedBy  ?symptom  ?part
WHERE {
        ?d a ppo:Disorder .
        ?d ppo:name ?disease .
        ?d ppo:has_case ?c .
        ?c ppo:has_plant_observation ?p .
        ?p ppo:observed_abnormality ?ab .
        ?ab ppo:value_category ?symptom .
        ?d ppo:has_factor ?f .
        ?f ppo:scientific_name ?causedBy .
        ?ab ppo:location ?loc .
        ?loc ppo:name ?part .
        FILTER CONTAINS(LCASE(?part),"stem").
         }
```

Execute

?disease	?causedBy	?symptom	
Gummosis^^xsd:string	Phytophthora spp.^^xsd:string	Lesions around circumference of trunk^^xsd:string	Stems^^xsd:string
Foot rot^^xsd:string	Phytophthora spp.^^xsd:string	Infection of young shoot near the ground^^xsd:string	Stems^^xsd:string

Fig.10. Diseases which cause abnormalities in Stem

SPARQL Query | Snap SPARQL Query

Snap SPARQL Query:

```
PREFIX xsd: <http://www.w3.org/2001/XMLSchema#>
PREFIX ppo: <http://www.ontologies.com/ppo#>
SELECT ?disease ?affected_plant_part ?symptom
WHERE {
        ?d a ppo:Disorder .
        ?d ppo:name ?disease .
        ?d ppo:has_case ?c .
        ?c ppo:has_plant_observation ?p .
        ?p ppo:observed_abnormality ?ab .
        ?ab ppo:value_category ?symptom .
        ?d ppo:has_factor ?f .
        ?f ppo:scientific_name ?causedBy .
        ?ab ppo:location ?loc .
        ?loc ppo:name ?affected_plant_part .
        FILTER CONTAINS(LCASE(?causedBy), "phytophthora") .

         }
```

Execute

?disease	?affected_plant_part	
Brown fruit rot^^xsd:string	Fruits^^xsd:string	Light brown, leathery areas on fruits^^xsd:string
Brown fruit rot^^xsd:string	Fruits^^xsd:string	Decay of fruits^^xsd:string
Brown fruit rot^^xsd:string	Fruits^^xsd:string	White fungus growth on fruit surface^^xsd:string
Root rot^^xsd:string	Roots^^xsd:string	Decaying of fibrous roots^^xsd:string
Root rot^^xsd:string	Roots^^xsd:string	Discolored and white thread-like roots^^xsd:string
Gummosis^^xsd:string	Stems^^xsd:string	Lesions around circumference of trunk^^xsd:string
Gummosis^^xsd:string	Bark^^xsd:string	Cracks with gum oozing out^^xsd:string
Foot rot^^xsd:string	Stems^^xsd:string	Infection of young shoot near the ground^^xsd:string
Seed Rot^^xsd:string	Seedlings^^xsd:string	Damping-off^^xsd:string

Fig.11. Diseases and symptoms caused by Phytophthora spp.

are triggered on the inferred model, to retrieve the inferences like disease occurrence and preventive measures. The framework developed in JAVA is generic and can be reused for any application or domain.

The user interface for the application is designed using JAVA Swing (Figure 12). The sensor values indicating different parameters from SenseQube are displayed and can be changed. The ”Fetch Inferences” button calls the semantic application framework and the model is created with the provided values as sensor observations and plant parameters. Selected SPARQL query is executed on the inferred model and the results are displayed in the panel (Figure 12).

5 Conclusion

The sensors and sensor data have immense potential to enhance the quality of our living. They have an impact in different application domains. The heterogeneity in the sources of information makes it necessary to implement unambiguous ontologies which clearly defines the source and meaning of the data in its respective domains. Semantics-driven approaches are needed to enable the reasoning engines to derive significant events from the voluminous IoT data generated. Semantic middle-ware is proposed which converts the IoT raw data to machine

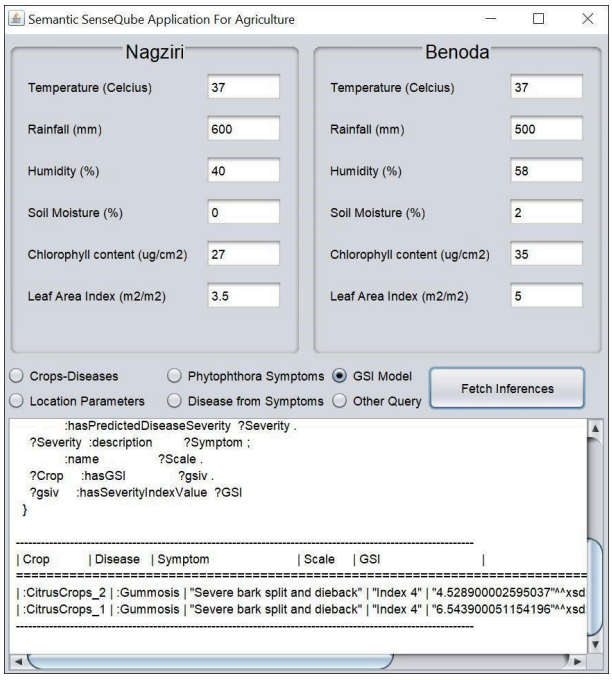


Fig.12. Semantic SenseQube Application User Interface

understandable models. Ontologies form the backbone of the semantic approach, by defining the terminology and relationships.

Semantics-driven agricultural decision support application is designed to use semantically enriched sensors for disease diagnosis. W3C recommended SSN ontology is used for sensors. Crop and plant pathology related Plant Protection ontology (PPO) is designed. SSNAGRI ontology is developed for IoT applications related to agriculture. Rules for disease prediction, preventive measures were formulated. Reasoning was done using rule based engines for inferences. The interoperable semantic middle-ware is developed which makes the domain knowledge explicit for the reasoning engines, thus making the IoT data machine understandable.

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